

Signal Processing: Methods and Algorithms

Laboratory 3

1 Wiener process

Consider the Wiener process $W(t)$ defined by

$$\frac{dW(t)}{dt} = F(t)$$

where $W(0) = 0$, and $F(t)$ is a white Gaussian noise with zero mean and autocorrelation function $R_F(t_1, t_2) = \delta(t_1 - t_2)$. Do the following:

1. Generate $M = 5$ realizations of the discrete-time Wiener process $W[n]$. Each realization must be made by $N = 1000$ samples, with a sampling time $T_s = 1$. Plot the realizations on the same figure. **Hint:** To generate the Wiener process, write a Matlab function whose inputs are the number of samples N and the sampling time T_s , and whose output is the simulated realization of the Wiener process $W[n]$.
2. By simulating a large number of realizations, estimate the probability density $f_W(x, t)$ of the Wiener process at some time instants chosen at will. Use the function `hist` to estimate the probability density.
3. Plot the estimated probability densities on the same figure, to highlight the increase in variance of the Gaussian probability density function of $W(t)$.

2 The transient of the Langevin equation

Consider the Langevin equation

$$\frac{dV(t)}{dt} + \beta V(t) = F(t)$$

where $V(0) = 0$ and $F(t)$ is a white Gaussian noise with zero mean and autocorrelation function $R_F(t_1, t_2) = \delta(t_1 - t_2)$. The solution $V(t)$ of the Langevin equation is the *Ornstein-Uhlenbeck process*. Do the following:

1. Generate $M = 50$ realizations of the Ornstein-Uhlenbeck process of $N = 100$ samples each, with $\beta = 0.01$ and $T_s = 1$. Plot the realizations in the same figure, and superimpose the two curves $\pm\sigma_V(t)$, where $\sigma_V(t)$ is the theoretical standard deviation of $V(t)$. **Hint:** To solve the Langevin equation, write a function whose input values are N , β , T_s , and whose output is the discrete-time Ornstein-Uhlenbeck process $V[n]$.
2. Generate $M = 5$ realizations of the Ornstein-Uhlenbeck process by changing β between two arbitrary values β_1 and β_2 . Choose a large β_1 and a small β_2 , to highlight the transition from the Ornstein-Uhlenbeck process to the Wiener process, which corresponds to $\beta = 0$.